

Muhammad Kamil Khan

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Portfolio: kamilkhan23.github.io/kamilkhan.github.io

Research Interests

Extended reality (XR); machine learning and reinforcement learning for interactive 3D systems and intelligent agents; 3D Gaussian Splatting and real-time rendering on standalone hardware; physics-based interaction and human factors of XR.

Education

Bachelor of Science in Computer Systems Engineering

Oct 2022 – Jul 2026

University of Engineering and Technology (UET) Peshawar

Peshawar, Pakistan

- GPA: 3.61 / 4.00
- Relevant coursework: Data Structures and Algorithms, Operating Systems, Systems Programming, Microprocessor-Based Systems, Digital Signal Processing, Artificial Intelligence, Game Development

Publications

- M. K. Khan and A. Bhojan. Physics-Aware Gaussian Splatting Interaction in Standalone Virtual Reality. Manuscript in preparation for IEEE AIVR.

Research Experience

Research Intern, Summer Research Internship (SRI 2026)

Jul 2026 – Sep 2026

School of Computing, National University of Singapore (Supervisor: Dr. Anand Bhojan)

Singapore

- Selected as 1 of 23 participants worldwide for the NUS School of Computing Summer Research Internship.
- Built a standalone VR system for physics-interactive manipulation of 3D Gaussian Splatting objects that runs natively on Meta Quest 3 without PC tethering, where previously reviewed systems were PC-tethered or rendering-only.
- Designed and implemented a point-cloud-based convex-hull collision generation algorithm, 50–150x faster than standard voxelization-based approaches.
- Designed a counterbalanced human-subjects VR study (IRB submitted) on physical interaction fidelity, using NASA-TLX and presence questionnaires.
- Reviewed literature on 3D/4D Gaussian Splatting and mobile and XR Gaussian rendering; presented progress in weekly research meetings.

Research Project, Cross-View Correspondence for Road-Image Homography

Jan 2026 – Feb 2026

Remote, with Dr. Fawad Ahmad, Rochester Institute of Technology

- Investigated automatically finding four corresponding anchor points between side-view and top-down road images to estimate a homography.
- Experimented with feature matching (ORB, SIFT/FLANN, LoFTR with RANSAC), geometric cues (Hough and LSD line intersections, rectangle constraints), and SegFormer road segmentation.

Projects

ARDetect: Offline AR Learning App for Electronics (Final Year Project, 4-person team)

2025 – 2026

Tools: Unity AR Foundation, YOLOv8n, ONNX, Unity Sentis, Python, Google Colab

- Android app that recognizes electronic components (breadboard, LED, Arduino) through the phone camera and places matching 3D models in AR, with an offline chatbot that answers questions about each part.
- Owned the model side: merged datasets and trained a 3-class YOLOv8n detector in Colab (61 epochs), reaching mAP50 of 0.993 / 0.848 / 0.995 (breadboard / LED / Arduino).
- Built the deployment pipeline: exported the model to ONNX and ran real-time inference on-device through Unity Sentis, so detection works fully offline with no cloud dependency.

Procedural AI-Driven Chunked World System (Unity)

- Runtime chunk-based procedural environment generation with seeded worlds, difficulty modes, and NavMesh AI navigation across generated terrain.
- Code: github.com/KamilKhan23/Procedural-AI-Chunked-World | Video: youtube.com/watch?v=yVf1r7poS2M

Recognition Without Dialogue: Non-Verbal Agent System (Unity)

- Artificial agent that communicates awareness through gaze, timing, and proximity, with persistent memory shaping its behavior.
- Code: github.com/KamilKhan23/Recognition-Without-Dialogue | Video: youtube.com/watch?v=U3J7lswDd6k

Professional Experience

Media and Evaluation Lead

Jul 2025 – Jun 2026

Youth Development Institute

Peshawar, Pakistan

- Supported a province-wide skills development program in Khyber Pakhtunkhwa targeting about 60,000 beneficiaries; coordinated delivery across 8 cities for 7,000+ trainees in three months.
- Produced documentary content for GIZ, CISNAR, and BMZ, and managed field teams and the organization's social media across branches.

Game Developer (promoted from Game Development Intern)

Sep 2024 – Apr 2025

Metal Heart Game Studios

Peshawar, Pakistan

- Developed a 3D superhero mobile game for Android and iOS in Unity and C#: physics-based playable character, traversal and combat systems, new levels and interactive objects.
- Optimized real-time performance across mobile devices and revamped a 5+ year-old legacy game's UI and runtime systems for modern platforms.
- Produced cinematic cutscenes with Cinemachine and Timeline.

2D Game Development Intern

Aug 2023 – Sep 2023

Spark Game Studios

Islamabad, Pakistan

- Designed, built, and published a 2D racing game with AI opponents, checkpoints, and leaderboards to Google Play, with ad monetization.

Summer School Intern (Finance and Registration Lead)

May 2023 – Sep 2023

Centre for Learning Law and Business (CLLB)

Peshawar, Pakistan

- Led finance and registration for a 7-day multidisciplinary summer school, working with a team of 20 interns; hosted its business pitch workshop.

Leadership and Honors

Director Management, UET Computer Society

Nov 2022 – Jul 2025

- Led a 50-member team in a 250+ member society; organized IGNITE, UET's inter-university tech festival, and events with 300+ participants.

Game Development Lead, Google Developer Student Club (UET)

Jan 2024 – Jul 2026

- Mentored students building their first Unity prototypes.

Creative Lead, Coders Nexus

Dec 2023 – Jul 2025

- Ran a 2.5-hour webinar on game development careers for 35+ participants.

Honor: Prime Minister Appreciation Award, Green Youth Movement

Skills

Programming Languages: C#, C++, C, Python, MATLAB, MIPS Assembly, Bash

XR and Graphics: Unity, Unity AR Foundation, OpenXR, Meta Quest 3, 3D Gaussian Splatting, Physics Simulation, Blender, Godot

Computer Vision and ML: PyTorch, OpenCV, ONNX, Unity Sentis, YOLOv8, SegFormer, LoFTR, Google Colab, Neural Networks, Time-Series ML

Research Methods: Human-Subjects Study Design, IRB Protocols, NASA-TLX, Presence Questionnaires, JASP, jamovi

Tools: Git, Linux (Ubuntu), Android/Gradle Build Pipelines, Embedded Systems, Microcontrollers

Languages: English (native), Urdu, Pashto